

EXPLORATORY DATA ANALYSIS

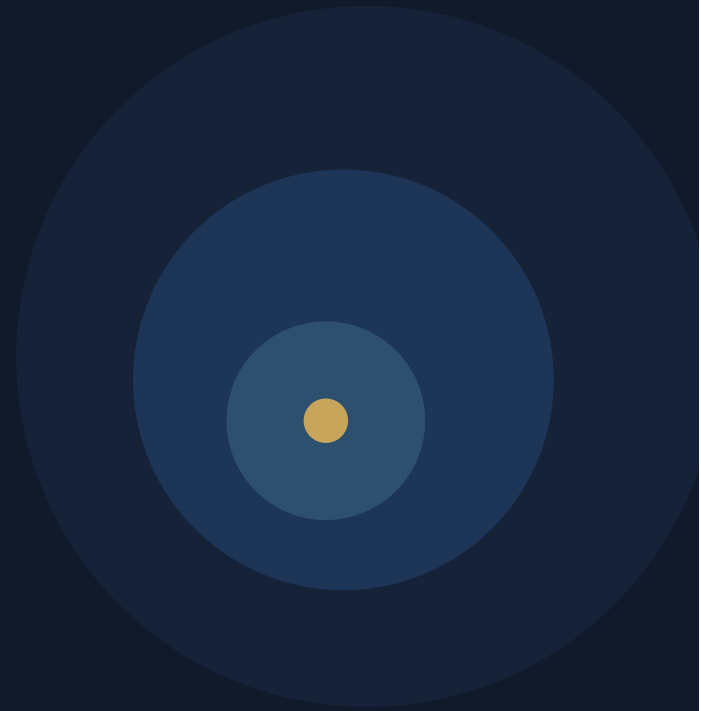
King County Housing Market

Finding the right home, in the right
neighborhood, at the right time for
Nicole Johnson.

Jishnu Vasudevan

Exploratory Data Analysis Project

AI ENGINEERING / DATA SCIENCE BOOTCAMP



CLIENT PROFILE

Nicole's requirements

What does the client want? Three things, in her own terms, before any data enters the room.

01

Neighborhood

Lively, central area.

02

Price range

Mid-market

03

Timing

Buying within the year

DATA FOUNDATION

Dataset and market overview

The evidence base for every finding that follows.

\$450k

Median sale price
across the dataset

Source

King County house sales

Coverage period

May 2014 – May 2015

Scope

21,597

sales records

22

original variables

Variables used

Price • Date • ZIP • Coordinates • Living space

FROM CLIENT NEEDS TO TESTABLE HYPOTHESES

Hypotheses based on Nicole's needs

Her brief, turned into three testable questions.

H1 Timing

BUY WITHIN THE YEAR

Homes sold in winter have a sale price at least 5% lower than in spring

TESTED ON SLIDE 05

H2 Centrality

STAY CENTRAL + LIVELY

Homes distant from center Seattle have a median sale price at least 25% lower than homes in the central Seattle

TESTED ON SLIDE 07

H3 Living space

PRICE + LOCATION

Does moving outward give more space?

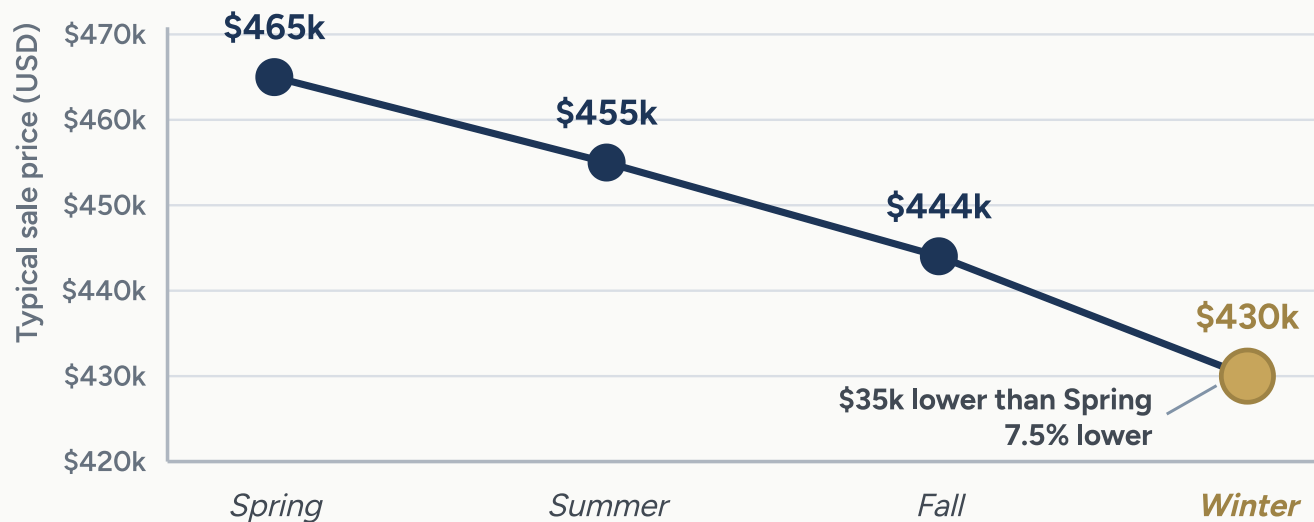
Within the middle-price range

TESTED ON SLIDE 08

H1 TIMING

When is the best time to buy?

Median sale price by season



FINDING

Winter has the lowest median price.

IMPLICATION FOR NICOLE

Potentially better prices, but fewer homes to choose from.

**Pattern within this one-year dataset.*

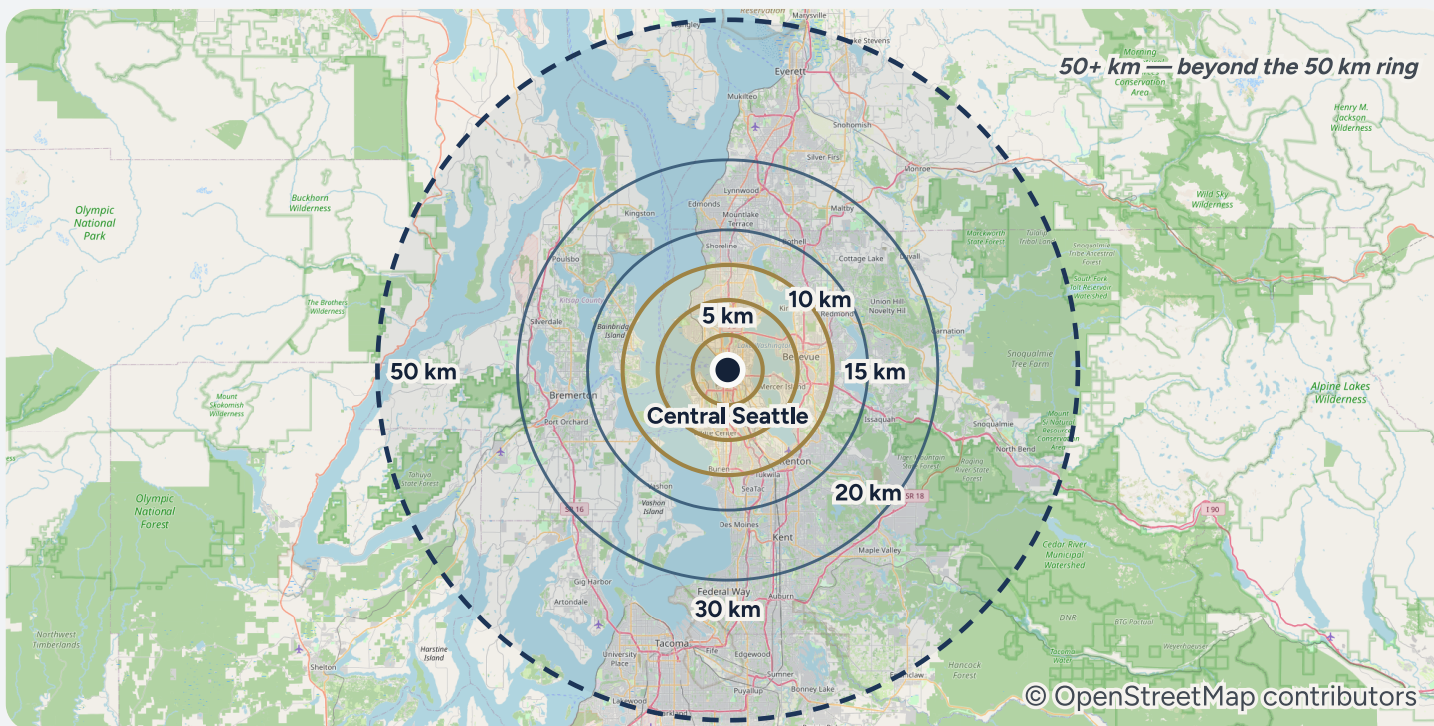
Typical price = median sale price.

Y-axis zoomed to make seasonal differences easier to read.

LOCATION CONTEXT

What do the different distance bands look like?

These are the distance bands used in the following analyses



DISTANCE BANDS

- 0–5 km
- 5–10 km
- 10–15 km
- 15–20 km
- 20–30 km
- 30–50 km
- 50+ km

HOW TO READ THIS

Inner bands = nearer Seattle

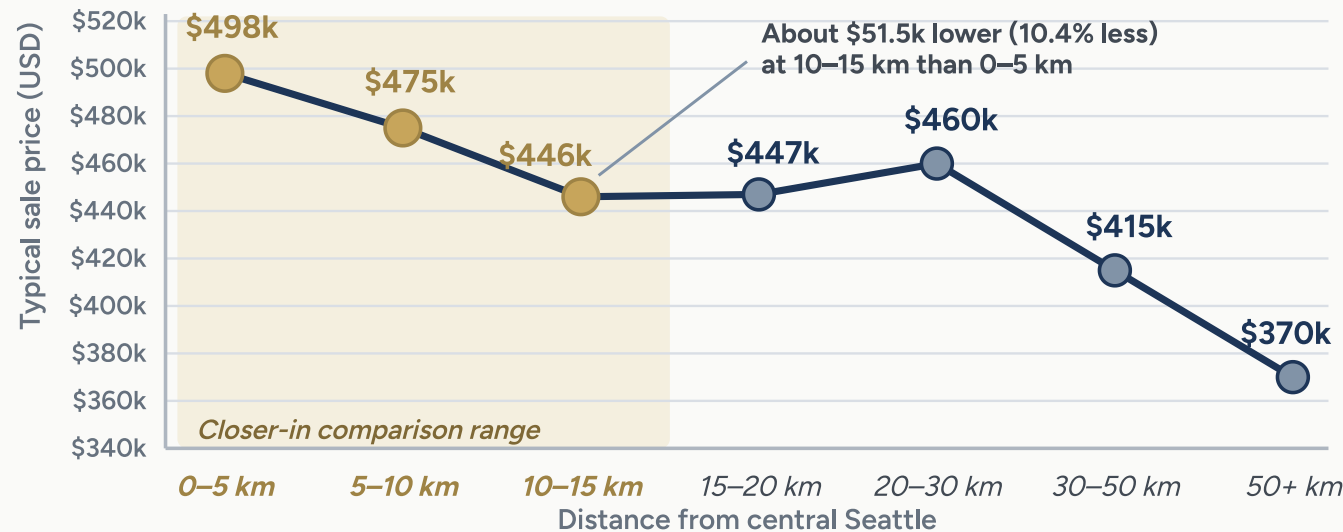
Outer bands = farther away

Straight-line distance from central Seattle — not travel time.

H2 CENTRALITY

How much does centrality cost?

Median sale price by distance band, 0–50+ km



FINDING

Central homes carry a price premium.

10–15 km $\approx$ 10% cheaper than 0–5 km

IMPLICATION FOR NICOLE

Looking slightly farther out may lower the price.

Typical price = median sale price.

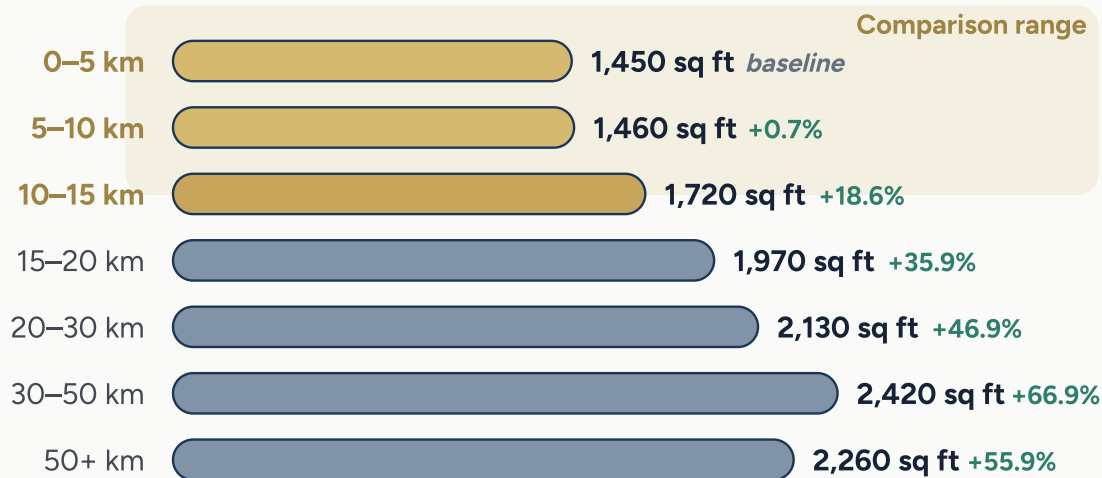
Shaded area is the 0–15 km comparison range.

Y-axis zoomed for readability.

H3 LIVING SPACE

What does she gain in living space?

Median living space by distance band, within her price range



FINDING

Living space generally increases farther from Seattle.

10-15 km = +19% space vs 0-5 km

IMPLICATION FOR NICOLE

More space within the same middle-price range.

*Analysis is restricted to homes within Nicole's middle-price range.

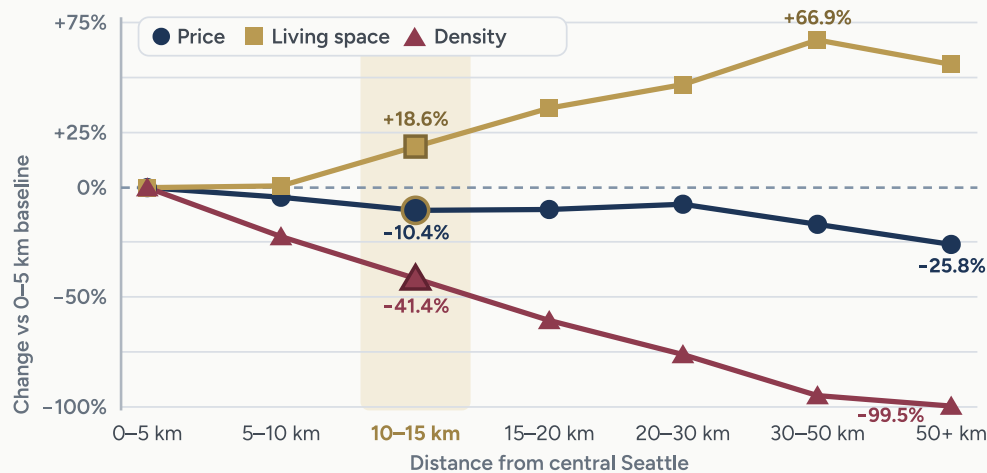
Typical space = median living space for homes within Nicole's middle-price range. Percentages show the increase vs the 0-5 km baseline.

Trade-offs

H1 + H2 + H3

Balancing price, space, and urban character

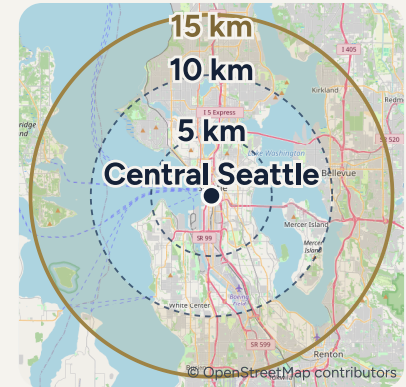
Moving farther out brings lower prices and more space, but also lower density



-10.4% price

+18.6% living space

-41% density



10-15 km remains relatively close to central Seattle

KEY TAKEAWAY

10-15 km offers a strong practical compromise.

*Density is used as a proxy for urban character.

Chips show the 10-15 km band vs the 0-5 km baseline. Below 0% = cheaper or less dense; above 0% = more living space.

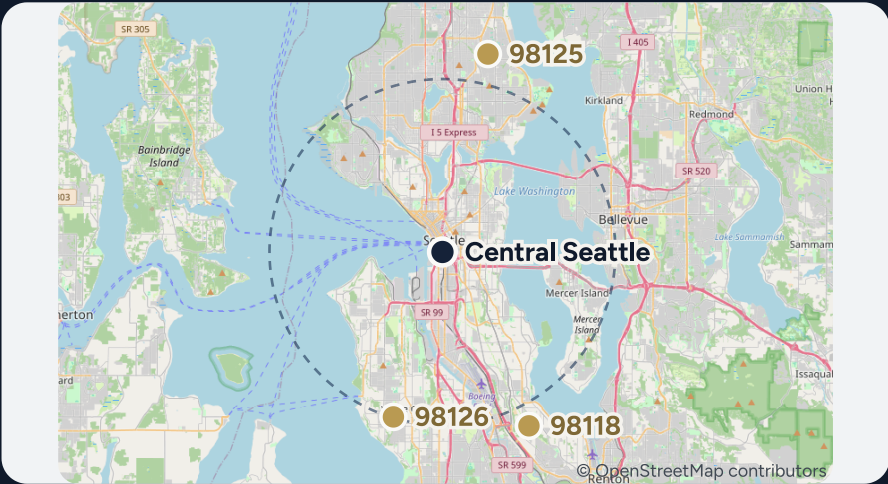
SHORTLIST

From screening to real ZIP codes

Three areas stood out as practical starting points

ANALYSIS → SCREENING → SHORTLIST

Screened on: price • space • density • distance • sales reliability



98118

Most living space

Price
\$420k

Distance
7.9 km

Space
1,740 sq ft

Sales
271

98126

Highest density

Price
\$420k

Distance
7.9 km

Space
1,560 sq ft

Sales
230

98125

Most suitable sales

Price
\$427k

Distance
12.0 km

Space
1,595 sq ft

Sales
308

Shortlist, not final ranking.

NEXT STEPS

Recommendations for Nicole

1

WHERE TO BUY

- Start with 98118, 98126, 98125
- Roughly 8–12 km from central Seattle

2

WHEN TO BUY

- Prefer winter if timing is flexible
- Lower prices, fewer choices

3

WHAT TO ADJUST

- Be flexible on exact centrality
- More space and lower price farther out

HOME FILTERS TO DECIDE NEXT



SPACE

Minimum sq ft



QUALITY

Grade / condition



AGE

Newer / older



LAYOUT

Beds / baths



FEATURES

View / balcony

LOOKING AHEAD

Limitations and next steps

| What this analysis cannot answer yet — and what to do next

LIMITATIONS

Historical data

2014–2015 only

One-year view

Seasonality is uncertain

Density proxy

Not direct liveliness data

NEXT STEPS

Update market data

Current prices and listings

Add location context

Transit, amenities, safety

Compare real listings

Against her own filters

Use the EDA to narrow the search, then validate with current data.

Thank you

Questions?

Jishnu Vasudevan

King County Housing Market — EDA Project

